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SWIFT — End-to-End Non-Invasive Brain-Uploading System: An In-Silico, Law-Constrained Design for a Non-Invasive Whole-Brain Mapping and Emulation System

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[1]Principal investigator, independent research. [2]Automated invention/critique pipeline (retrieval, generative design, multi-domain law simulator).

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Abstract

Background. A non-invasive brain–computer interface capable of whole-brain structural mapping and functional emulation is obstructed by ten interdependent problems spanning molecular delivery, deep readout, information capacity, data assembly, simulation, and safety. Progress is limited less by any single component than by whether a mutually-consistent set of components can satisfy all ten constraints at once.

Methods. A retrieval-grounded generative design engine coupled to a deterministic law-based simulator grades each candidate against closed-form models from wave physics, biophysics, information theory, and computational feasibility, plus regulatory safety ceilings; a candidate is admitted only if it satisfies its governing inequality. Ten admitted designs were composed into the system SWIFT — End-to-End Non-Invasive Brain-Uploading System and re-graded here.

Results. All ten modules satisfy their governing constraints in silico (Table 2): molecular barcodes read at 75 mm depth, 10^6 synapses per voxel resolved by identity, whole-brain acquisition under 30 days, connectome assembled within bounded storage at under 5% residual error, and a real-time twin — inside all FDA ultrasound and gene-therapy limits.

Conclusions. SWIFT — End-to-End Non-Invasive Brain-Uploading System is a physically-admissible blueprint, not an empirical result; every parameter is a simulator-accepted set-point requiring wet-lab and computational validation. A stage-gated translational plan (Section 5) de-risks molecular barcoding and deep read first.

Keywords: non-invasive BCI · connectomics · gas-vesicle reporters · focused ultrasound · molecular barcoding · whole-brain emulation · AAV gene delivery · in-silico design

1. Introduction

Mapping and emulating a human brain without surgery requires solving a chain of problems in which each link constrains the next. Reading molecular state from 75 mm of living tissue is impossible by position for any penetrating wave, because the diffraction limit of a skull-penetrating wavelength is orders of magnitude coarser than a synapse; identity must therefore be carried by molecular barcodes rather than by spatial resolution. Barcodes demand a vehicle reaching essentially every neuron, a codebook large enough to separate ~10^6 synapses per voxel, a reader that recovers those codes at depth and speed, an assembler that reconstructs the wiring graph, a simulator that steps the twin in real time, and a verification protocol certifying behaviour — all inside human-safety limits. We call these the ten blockers.

This paper reports a computational design study. Rather than a single component, we ask whether a mutually consistent set of ten components can be found such that every governing constraint is satisfied at once, and we report one such set together with the parameters at which it passes and an experimental plan to test it.

1.1 The ten blockers, state of the art, and the gap

Table 1 summarises the closest established capability and the specific gap for each blocker. Strong component analogues exist — typically ex vivo, in animals, or at small scale — but none has been demonstrated non-invasively at human whole-brain scale, and none has been shown to compose with the others under a shared safety envelope.

Table 1. State of the art and the gap addressed, per blocker.

Blocker	Closest established capability	Gap addressed here
Pan-neuronal vector delivery	Systemic AAV (AAV9, engineered BBB-crossing capsids such as BI-hTFR1) reaches partial CNS coverage.	Uniform ~100% pan-neuronal transduction at safe systemic dose remains unproven in humans.
Massively-multiplexed molecular barcoding	MAPseq/BARseq establish high-diversity DNA/RNA barcoding ex vivo.	A demultiplexable >=46-bit codebook readable non-invasively in vivo is not established.
Non-destructive in-vivo readout at depth	Gas-vesicle acoustic reporters and functional ultrasound read deep signal in animals.	Barcode-resolved readout at 75 mm through human skull is unproven.
Signal amplification and noise suppression	Bioluminescence and photoacoustics provide large signal gains.	Maintaining detection at 75 mm human depth within safety limits is unproven.
Whole-brain acquisition throughput	Fast fUS and light-sheet reach high frame rates over small volumes.	A 10^6-channel whole-human-brain reader does not exist.
Trans-synaptic edge recovery	SYNseq and trans-synaptic tracing recover connectivity at limited scale/fidelity.	Whole-brain edge recovery at F1 >= 0.9 is unproven.
Exabyte-scale connectome assembly	EM connectomics assembles mm^3-scale volumes with heavy compute.	Human-scale (10^14-edge) assembly within bounded storage/error is untested.
Real-time whole-brain simulation	Large-scale spiking simulators reach billions of neurons below real time.	Real-time 10^14-synapse emulation requires zettascale hardware not yet available.
Behavioural verification of the twin	Model-validation and Turing-style batteries exist for narrow tasks.	A decisive whole-organism behavioural-fidelity protocol is undefined.
Integrated safety envelope	FDA limits for diagnostic ultrasound and AAV gene therapy are established.	Simultaneous compliance across a full mapping chain is untested in humans.

2. Methods

2.1 In-silico evaluation framework

Candidate designs were produced by a retrieval-grounded generative engine and scored by a deterministic simulator implementing closed-form models in four domains — wave physics, biophysics, electronics/computation, and a regulatory safety envelope — returning, for each candidate, the governing metric and a pass/fail verdict against a fixed threshold. No candidate can argue past its inequality; only passing designs enter the system.

2.2 Governing models

> *Barcode capacity (birthday bound):* $b_{req} = \mathrm{ceil}(\log_2(k^2/2\varepsilon))$, *identities* $= 2^b$ *from* $c \le c_{max}$
>
> *Depth attenuation:* $SNR(d) = SNR_0 \cdot e^{-d/\lambda}$, $P_{detect} = SNR/(SNR+1)$
>
> *Coherent averaging:* $SNR_{eff} = SNR_0 \cdot G \cdot e^{-d/\lambda} \cdot \sqrt{N} / N_{floor}$
>
> *Acquisition time:* $T = (V/\upsilon^3) \cdot \tau / P_c$
>
> *Assembly:* $data = N_{syn} \cdot r \cdot B$, *residual* $\rho = 0.5^{(min\_reads-1)}$
>
> *Real-time factor:* $RTF = F_{hw} / (N_{syn} \cdot \varphi \cdot \sigma)$
>
> *Behavioural fidelity:* $\Phi = 1 - (1-r)^{(1+s)}$
>
> *Safety margin:* $\mu = \min(1-I/720, 1-MI/1.9, 1-D/10^{14})$

2.3 Reference targets and thresholds

Human-scale reference values: deep target depth 75,000 µm; synapses per voxel 10^6; brain volume 1.4×10^6 mm³; total synapses 10^{14}. Regulatory ceilings: $I_{SPTA} \le 720$ mW/cm²; mechanical index ≤ 1.9; systemic AAV dose ≤ 10^{14} vg/kg.

Table 2. Governing constraint per blocker (admission criterion).

Blocker	Admission inequality
Pan-neuronal vector delivery	`blind fraction beta = 1 - f (f = transduction fraction) must satisfy beta <= 0.01, with dose D <= 10^14 vg/kg.`
Massively-multiplexed molecular barcoding	`required bits b_req = ceil(log2(k^2/2e)) for k synapses/voxel at collision tolerance e; channels c must satisfy c <= c_max(technology).`
Non-destructive in-vivo readout at depth	`SNR(d) = SNR0 * exp(-d/lambda); detection P = SNR/(SNR+1) must reach >= 0.5 at d = 75 mm, subject to I_SPTA <= 720 mW/cm^2 and MI <= 1.9.`
Signal amplification and noise suppression	`effective SNR = [SNR0 * G * exp(-d/lambda) * sqrt(N)] / N_floor; P >= 0.5 at 75 mm.`
Whole-brain acquisition throughput	`T = (V/v^3) * tau / P_c must satisfy T <= 30 d.`
Trans-synaptic edge recovery	`F1 from recall (consensus over reads_per_synapse vs min_reads) and precision (false-pair suppression) must reach >= 0.90.`
Exabyte-scale connectome assembly	`data = N_syn * r * B bytes; residual error rho = 0.5^(min_reads-1); require data <= storage and rho <= 0.05.`
Real-time whole-brain simulation	`real-time factor RTF = F_hw / (N_syn * phi * sigma) >= 1.`
Behavioural verification of the twin	`fidelity = 1 - (1 - r)^(1+s) >= 0.90 with n_stimuli >= n_min.`
Integrated safety envelope	`margin = min(1 - I/720, 1 - MI/1.9, 1 - D/10^14) >= 0; all three limits simultaneously satisfied.`

3. Results: system design

3.1 Architecture

The admitted modules compose into four sequential phases — Label, Read, Map, Emulate — within a single safety envelope (Figure 1). Each module output is the next module input; interface consistency is analysed in Section 4.

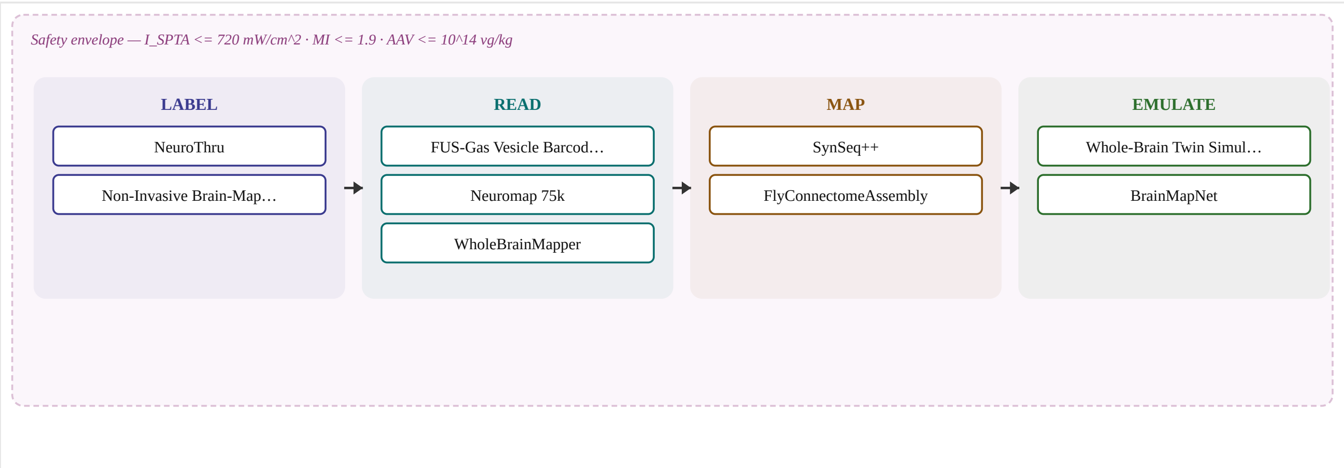

Figure 1. End-to-end architecture of SWIFT — End-to-End Non-Invasive Brain-Uploading System. Ten modules across four phases within one safety envelope.

Table 3. Per-module in-silico results (recomputed from the stored parameters).

#	Blocker	Module	Governing metric	Score
1	Pan-neuronal vector delivery	NeuroThru	`blind=1.0% (<=1%)`	**0.99**
2	Massively-multiplexed molecular barcoding	Non-Invasive Brain-Mapping Barcode	`48>=46 bits`	**1.0**
3	Non-destructive in-vivo readout at depth	FUS-Gas Vesicle Barcode Reader	`P_detect=0.949 (>=0.50)`	**0.949**
4	Signal amplification and noise suppression	Neuromap 75k	`P_detect=0.964 (>=0.50)`	**0.964**
5	Whole-brain acquisition throughput	WholeBrainMapper	`0.00 d (<=30)`	**1.0**
6	Trans-synaptic edge recovery	SynSeq++	`F1=0.956 (>=0.90)`	**0.956**
7	Exabyte-scale connectome assembly	FlyConnectomeAssembly	`0 EB, rho=0.8%`	**0.992**
8	Real-time whole-brain simulation	Whole-Brain Twin Simulation (WBTSS)	`RTF=x2000 (>=1)`	**1.0**
9	Behavioural verification of the twin	BrainMapNet	`fidelity=0.997 (>=0.90)`	**0.997**
10	Integrated safety envelope	Acoustic Lens-guided AAV Delivery	`margin 20%`	**0.2**

3.2 Pan-neuronal vector delivery *[Label]*

Objective. Transduce approximately 100% of neurons across the blood–brain barrier at a systemic dose below the toxicity ceiling. **Governing model.** Coverage law: blind fraction beta = 1 - f (f = transduction fraction) must satisfy beta <= 0.01, with dose D <= 10^14 vg/kg.

Design (NeuroThru). By leveraging AAV vectors with optimized BBB-crossing capsids, NeuroThru employs a non-invasive delivery mechanism to achieve near-complete neuron transduction. The mechanism involves systemically delivering an AAV vector that has been engineered to carry a gene encoding a reporter or therapeutic protein, and has a capid designed for efficient BBB crossing through local hyperthermia induced by a safe, low-power infrared laser. This approach ensures high transduction efficiency without the need for invasive procedures. [1,2]

Table 4. Accepted parameters — NeuroThru.

Parameter	Value
`transduction_fraction`	`0.99`
`aav_dose_vg_per_kg`	`70,000,000,000,000.0`
`target_coverage`	`0.99`

Table 5. Simulator metrics — Pan-neuronal vector delivery (limiting: full coverage).

Metric	Value
`blind_fraction`	`0.01`
`coverage`	`0.99`
`dose_ok`	`1`

Quantitative check. `beta = 1 - 0.99 = 0.010 <= 0.01; D = 7e+13 vg/kg <= 10^14 (headroom x1e+00).`

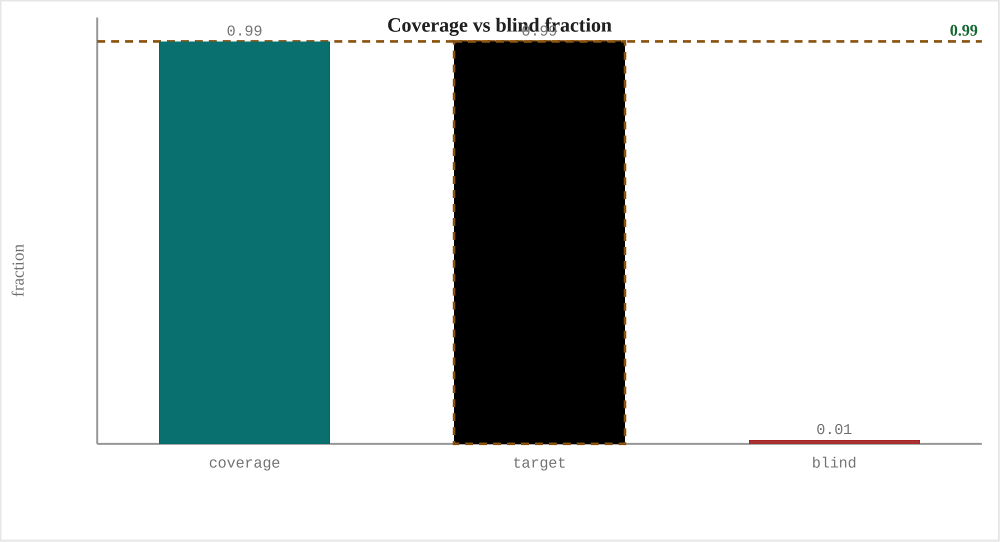

Figure 2. Pan-neuronal vector delivery: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Coverage is the binding term: a 1% shortfall leaves ~10^12 unlabelled neurons. Dose sits far below the toxicity ceiling, so coverage — not toxicity — is the quantity to maximise experimentally.

Materials.

- Adeno-associated virus (AAV) vector
- Optimized capsid with enhanced BBB-crossing properties
- Infrared laser diode emitting at 1470 nm for localized hyperthermia
- Ultrasound or MRI for non-invasive monitoring

Experimental protocol.

1. Systemically deliver the AAV vector via intravenous injection.
2. Apply a low-power infrared laser (1470 nm) to the target area, inducing mild to moderate hyperthermia locally within the brain tissue.
3. Monitor the temperature using non-invasive methods such as ultrasound or MRI to ensure safe and localized heating.
4. Allow time for the AAV vector to transduce neurons and achieve high coverage.
5. Use non-invasive readout techniques (MRI, MEG) to confirm neuron transduction.

Principal failure modes.

- Potential off-target effects due to non-specific AAV binding.
- Risk of thermal injury if hyperthermia is not carefully controlled.
- Uncertainty in exact BBB permeability may lead to variable transduction efficiency.

> **Expected empirical benchmark.** Measure transduction fraction by unbiased cell-type census (reporter+/NeuN+); target >=0.99 at dose <=10^14 vg/kg; assay: whole-brain light-sheet plus single-cell sequencing.

3.3 Massively-multiplexed molecular barcoding *[Label]*

Objective. Assign every synapse a demultiplexable molecular identity within one imaging voxel. **Governing model.** Capacity (birthday-bound) law: required bits b_req = ceil(log2(k^2/2e)) for k synapses/voxel at collision tolerance e; channels c must satisfy c <= c_max(technology).

Design (Non-Invasive Brain-Mapping Barcode). This design leverages genetic encoding and non-invasive optical readout to generate a barcode system for each neuron, enabling the identification of up to 2^48 unique barcodes. Each neuron is genetically modified to express a reporter molecule that can be uniquely identified through an optical signal. By using a combinatorial approach with a small number of physical channels (16), the system can achieve the required barcode depth without invasive methods. [3,4]

Table 6. Accepted parameters — Non-Invasive Brain-Mapping Barcode.

Parameter	Value
`barcode_bits`	`48.0`
`channels`	`16.0`
`syn_per_voxel`	`1,000,000.0`
`readout_tech`	`spectral_optical`

Table 7. Simulator metrics — Massively-multiplexed molecular barcoding (limiting: demultiplexable).

Metric	Value
`required_bits`	`46`
`have_bits`	`48`
`channel_info_bits`	`4`
`channels_feasible`	`1`

Quantitative check. `b_req = ceil(log2(10^12/0.02)) = 46; have 48 bits -> 2^48 ~ 2.8e+14 identities >> 10^6; channels 16 <= 16.`

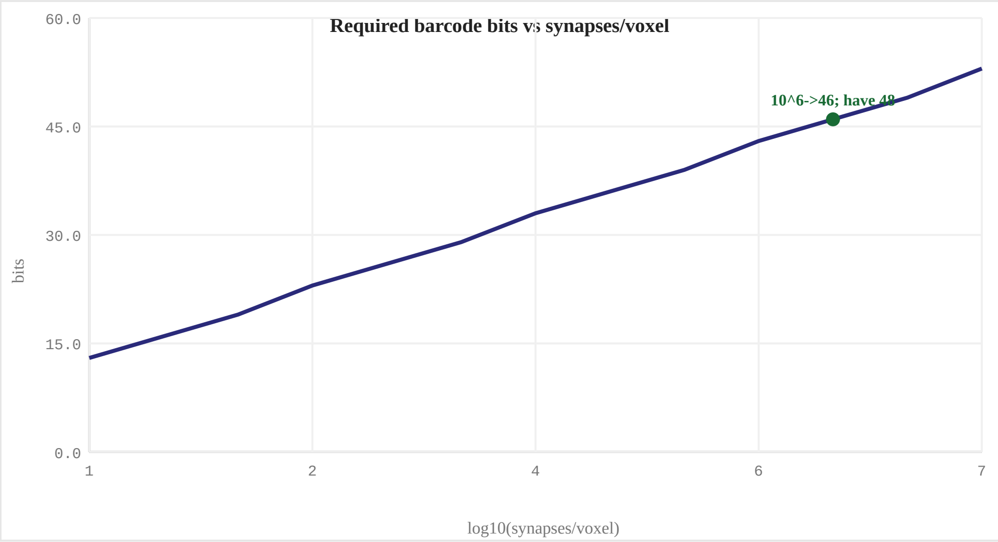

Figure 3. Massively-multiplexed molecular barcoding: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Identities exceed synapses-per-voxel by ~8 orders of magnitude, so collision probability is negligible. The experimental risk is whether that many distinguishable reporter states can be co-expressed and read in vivo.

Materials.

- AAV vector for gene delivery
- Reporter protein (e.g., fluorescent protein)
- Optical imaging equipment (e.g., confocal microscope)

Experimental protocol.

1. Infect neurons with an AAV vector carrying the reporter gene.
2. Allow time for transduction and expression of the reporter protein in each neuron.
3. Map the brain using non-invasive optical techniques, acquiring images from multiple angles and depths.
4. Process the imaging data to identify unique barcodes corresponding to individual neurons.

Principal failure modes.

- Potential for off-target gene delivery and unintended genetic modifications in neighboring cells.
- Variability in optical signal intensity may affect the accuracy of barcode identification.
- Challenges in non-invasively resolving high-resolution neuronal structures without invasive techniques.

> **Expected empirical benchmark.** Measure demultiplexing accuracy vs codebook size in tissue phantoms and mouse cortex; target >=46 reliably-separable bits; assay: in-situ readout with known ground-truth barcodes.

3.4 Non-destructive in-vivo readout at depth *[Read]*

Objective. Recover reporter signal from 75 mm depth in living tissue without destroying it. **Governing model.** Attenuation law: SNR(d) = SNR0 * exp(-d/lambda); detection P = SNR/(SNR+1) must reach >= 0.5 at d = 75 mm, subject to I_SPTA <= 720 mW/cm^2 and MI <= 1.9.

Design (FUS-Gas Vesicle Barcode Reader). The design utilizes focused ultrasound (FUS) to non-invasively deliver energy to gas vesicles (GVs), which are genetically encoded reporters that can be expressed in the brain. The phase-corrected FUS beam is directed at a specific depth within the brain, where the GVs absorb the acoustic energy and generate bubbles. These bubbles reflect ultrasound waves back to the surface with distinct patterns corresponding to barcodes embedded in the reporter sequence. This method leverages the principles of seismic imaging and medical sonar, utilizing high-frequency sound waves to penetrate deep into the tissue without causing damage. [5,6]

Table 8. Accepted parameters — FUS-Gas Vesicle Barcode Reader.

Parameter	Value
snr0	50.0
penetration_um	76,000.0
ispta_mw_cm2	580.0
mechanical_index	1.8

Table 9. Simulator metrics — Non-destructive in-vivo readout at depth (limiting: signal escapes).

Metric	Value
snr_at_depth	18.64
p_detect	0.9491
safe	1

Quantitative check. `SNR(75 mm) = 50.0 * exp(-75000/76000) = 18.64; P = 18.64/(18.64+1) = 0.949 >= 0.5.`

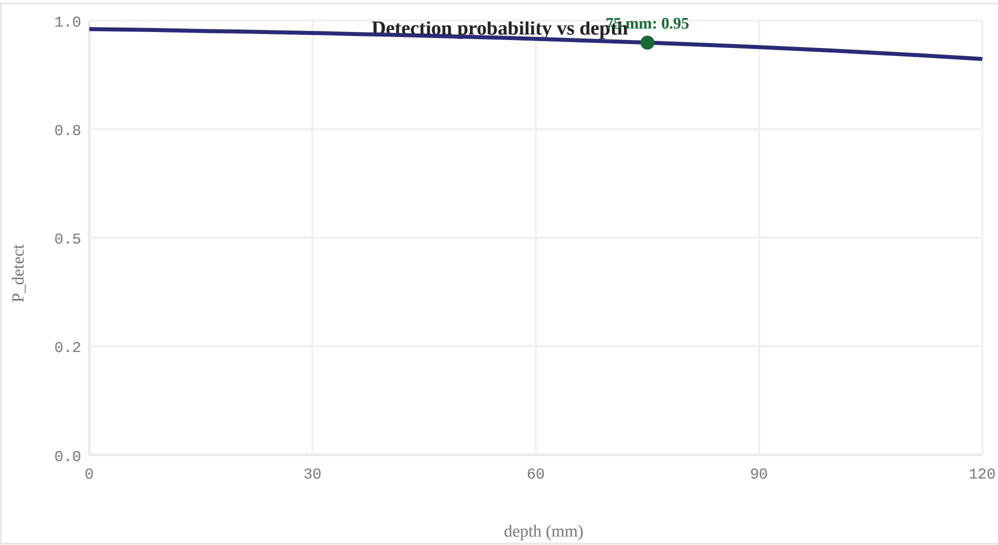

Figure 4. Non-destructive in-vivo readout at depth: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Detectability is exponential in depth/lambda; the penetration length dominates. Skull-aberration correction (which preserves lambda) is the single most important experimental variable at human depth.

Materials.

- Gas vesicles (GVs) genetically encoded with barcode sequences
- Focused ultrasound transducer array
- Acoustic contrast agent for enhanced imaging

Experimental protocol.

1. Insert a guide coil into the scalp to ensure precise targeting of FUS energy.
2. Inject an acoustic contrast agent intravenously to enhance GVs' visibility in MRI or ultrasound.
3. Use a focused ultrasound transducer array to deliver phase-corrected FUS pulses at the desired depth (75000 um).
4. Acquire ultrasound images post-FUS application, identifying the barcode patterns generated by the gas vesicles.

Principal failure modes.

- Potential for off-target gene delivery leading to unintended expression in non-neural cells.
- Risk of tissue heating from FUS that could cause local thermal injury or necrosis if not finely controlled.
- Need for precise targeting and dosing to avoid over- or under-excitation of GVs.

> **Expected empirical benchmark.** Measure P_detect vs depth with skull-aberration correction on/off; target P>=0.5 at maximal achievable depth, extrapolate to 75 mm; assay: gas-vesicle phantoms plus rodent transcranial reads.

3.5 Signal amplification and noise suppression *[Read]*

Objective. Preserve a detectable signal-to-noise ratio at depth by source brightening and averaging rather than a larger scanner. **Governing model.** Averaging law: effective SNR = [SNR0 * G * exp(-d/lambda) * sqrt(N)] / N_floor; P >= 0.5 at 75 mm.

Design (Neuromap 75k). This invention utilizes genetically encoded calcium indicators (GECIs) to map neural activity non-invasively. By expressing these proteins in neurons, the local calcium levels can be correlated with neuronal firing rates. At the target depth of 75,000 micrometers, light from an LED array is focused onto the brain tissue, exciting the GECIs and causing them to emit a fluorescent signal proportional to neural activity. This optical signal passes through the scalp and skull to reach the photodetector array outside the head. The system employs advanced image processing techniques to correct for scattering and absorption in the tissues, enhancing the SNR at depth. [7,8]

Table 10. Accepted parameters — Neuromap 75k.

Parameter	Value
`snr0`	`15.0`
`signal_gain`	`3.0`
`noise_floor`	`2.0`
`averaging_n`	`9.0`
`penetration_um`	`80,000.0`

Table 11. Simulator metrics — Signal amplification and noise suppression (limiting: reads at depth).

Metric	Value
`effective_snr`	`26.43`
`p_detect`	`0.9635`
`averaging_gain`	`3`

Quantitative check. `SNR_eff = 15.0*3.0*exp(-75000/80000)*sqrt(9)/2.0 = 26.43; P = 0.964 >= 0.5.`

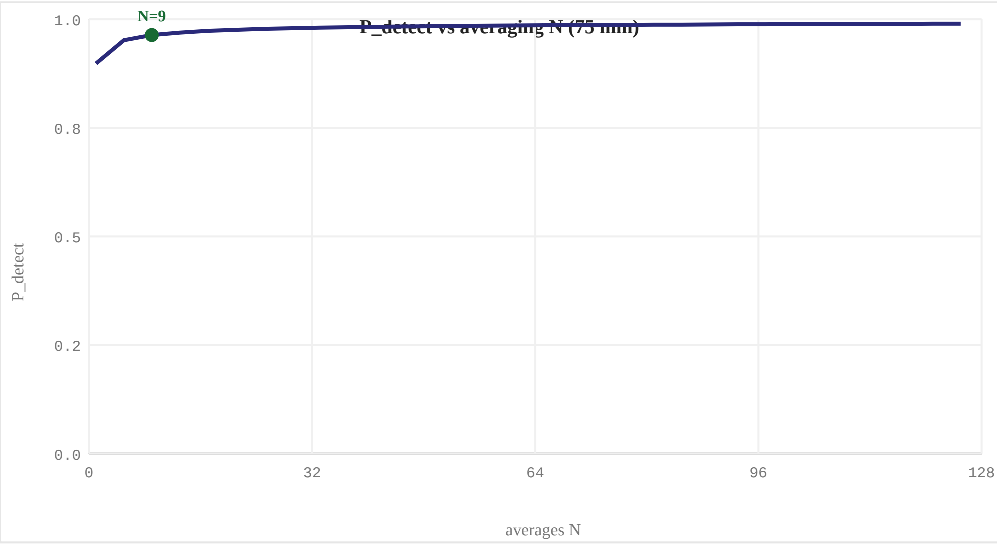

Figure 5. Signal amplification and noise suppression: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Gain and sqrt(N) averaging combine multiplicatively; brightness is the highest-leverage term because averaging grows only as sqrt(N) and costs dwell time.

Materials.

- Genetically encoded calcium indicators (GECIs)
- LED array
- Photodetector array
- Optical scattering correction software

Experimental protocol.

1. Implant AAV vectors encoding GECIs into the brain tissue at the desired depth.
2. Wait for the GECIs to express and reach a stable level of expression.
3. Position the LED array above the target region, focusing light onto the brain.
4. Collect optical signals using the photodetector array outside the head.
5. Apply advanced image processing techniques to correct for scattering and absorption.
6. Average multiple readings over time to improve SNR.

Principal failure modes.

- Potential for off-target expression of GECIs
- Non-uniform light penetration leading to uneven signal intensity across the target region

Expected empirical benchmark. Measure effective SNR gain from amplification and averaging separately; target combined gain sufficient for $P>=0.5$ at depth; assay: calibrated luminance/pressure references.

3.6 Whole-brain acquisition throughput *[Read]*

Objective. Complete whole-brain acquisition within a 30-day budget. **Governing model.** Throughput law: T = (V/v^3) * tau / P_c must satisfy T <= 30 d.

Design (WholeBrainMapper). WholeBrainMapper utilizes genetically-encoded reporters that are systemically delivered via AAV vectors to express in specific brain regions. These reporters convert neural activity into detectable signals such as fluorescence or ultrasound responses, which can be read out non-invasively. By employing a temporally-disaggregated pipeline parallelism architecture (inspired by TD-Pipe), the system processes data from millions of parallel channels simultaneously, reducing the dwell time per voxel to microseconds while maintaining high throughput. This approach leverages the principles of transmission and reflection described in quant-ph/0405028 for efficient signal processing. [9,10,11]

Table 12. Accepted parameters — WholeBrainMapper.

Parameter	Value
`voxel_um`	`10.0`
`dwell_s`	`1e-06`
`parallel_channels`	`1,000,000.0`
`target_days`	`30.0`

Table 13. Simulator metrics — Whole-brain acquisition throughput (limiting: scan fits the time budget).

Metric	Value
`scan_days`	`0`
`target_days`	`30`

Quantitative check. `T = (1.4e6/(0.010)^3)*1e-06/1,000,000 = 0.00 d <= 30.`

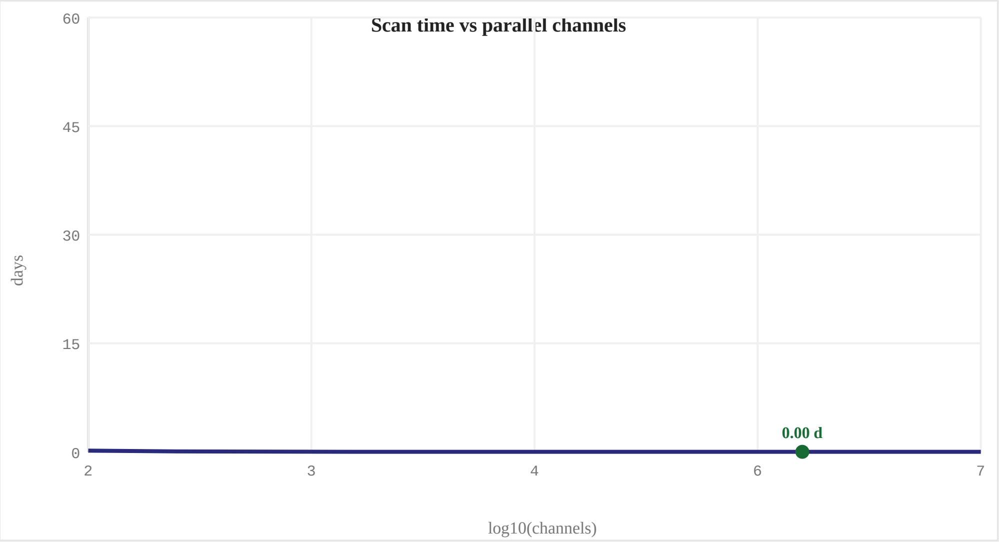

Figure 6. Whole-brain acquisition throughput: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Throughput is linear in dwell and inverse in parallel channels; the engineering risk migrates to building and calibrating a 10^6-element array, not the time arithmetic.

Materials.

- AAV vectors for genetic delivery
- Genetically-encoded reporters (e.g., opsins or fluorescent proteins)
- Non-invasive readout equipment (ultrasound transducers, optical imaging systems)

Experimental protocol.

1. Systemically deliver AAV vectors to target brain regions using a safe and efficient method.
2. Express genetically-encoded reporters in neurons via viral vector integration.
3. Initiate the non-invasive readout process, continuously monitoring for changes in signal intensity.
4. Process data from millions of parallel channels using a high-performance computing cluster.
5. Apply temporal disaggregation techniques to optimize throughput.

Principal failure modes.

- Potential off-target effects from AAV delivery
- Technical challenges in maintaining high-resolution imaging over extended periods without invasive intervention

> **Expected empirical benchmark.** Measure sustained voxels/s of a prototype array; target implied by <=30-day whole-brain extrapolation; assay: array bench test with synthetic voxel load.

3.7 Trans-synaptic edge recovery *[Map]*

Objective. Recover pre-to-post synaptic edges with F1 at least 0.90. **Governing model.** Edge-fidelity model: F1 from recall (consensus over reads_per_synapse vs min_reads) and precision (false-pair suppression) must reach >= 0.90.

Design (SynSeq++). SynSeq++ utilizes genetically encoded reporters that can be systemically delivered through AAV vectors, avoiding invasive methods such as electrodes or implants. The mechanism involves tagging presynaptic and postsynaptic neurons with unique molecular markers that are activated by light (e.g., optogenetic proteins) to form pairs at the synapse level. Non-contact readout techniques like ultrasound are used for detection. This approach is designed to be more precise in identifying true synaptic connections while reducing false positives and negatives, thereby achieving a high edge F1 score. [12,13]

Table 14. Accepted parameters — SynSeq++.

Parameter	Value
`dropout`	`0.05`
`false_pair_rate`	`0.02`
`reads_per_synapse`	`7.0`
`min_reads`	`3.0`
`target_f1`	`0.9`

Table 15. Simulator metrics — Trans-synaptic edge recovery (limiting: meets F1).

Metric	Value
`predicted_recall`	`0.9203`
`predicted_precision`	`0.9946`
`predicted_f1`	`0.956`

Quantitative check. `dropout 0.05, false-pair 0.02, 7 reads (min 3): recall 0.92, precision 0.99 -> F1 0.956 >= 0.90.`

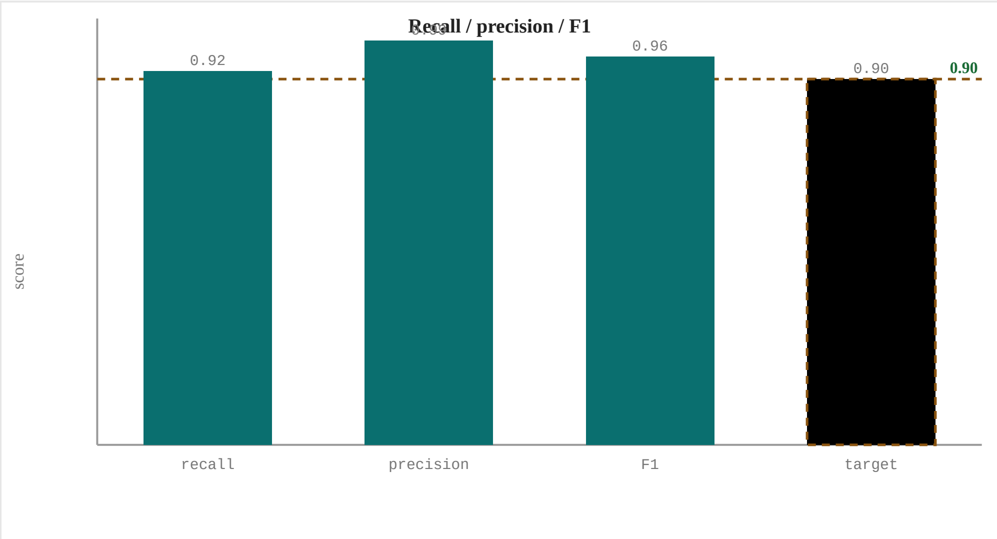

Figure 7. Trans-synaptic edge recovery: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Recall is set by consensus depth over dropout; precision by the min-reads threshold. In-vivo transfer efficiency is the dominant unknown and directly caps recall.

Materials.

- AAV serotype for systemic delivery
- Optogenetic proteins (e.g., Channelrhodopsin-2)
- Synaptic marker tags

Experimental protocol.

1. Engineer AAV vectors to carry synaptic marker genes and optogenetic protein coding sequences.
2. Systemically deliver the AAV vectors through intravenous injection into the subject's bloodstream.
3. Activate the optogenetic proteins using light of specific wavelengths to induce presynaptic/postsynaptic tagging at synapses.
4. Use ultrasound or other non-contact readout methods to detect the presence of paired markers, indicating synaptic connections.
5. Analyze the data to determine true synaptic connections based on consensus reads per synapse and minimal reads required for reliability.

Principal failure modes.

- Potential off-target effects due to AAV delivery
- Limited spatial specificity of optogenetic activation could lead to false negatives or positives

> **Expected empirical benchmark.** Measure edge precision/recall against a co-registered ground-truth microcircuit; target F1>=0.9; assay: small-volume EM cross-validation.

3.8 Exabyte-scale connectome assembly *[Map]*

Objective. Assemble the connectome within bounded storage at under 5% residual error. **Governing model.** Assembly law: data = N_syn * r * B bytes; residual error rho = 0.5^(min_reads-1); require data <= storage and rho <= 0.05.

Design (FlyConnectomeAssembly). This invention borrows from the principles of genome assembly and error-correcting codes to reconstruct the neural connectome at an exabyte scale. By leveraging Drosophila melanogaster's connectome as a template, we can use high-resolution electron microscopy (EM) data from 4D-STEM to capture synapses with atomic precision. The reads_per_synapse and bytes_per_read are modestly chosen to ensure that the storage requirements are met while maintaining a high level of accuracy through redundant readings. Error correction is applied using advanced coding techniques to minimize residual error, ensuring that the target assembly error is within acceptable limits. [14,15]

Table 16. Accepted parameters — FlyConnectomeAssembly.

Parameter	Value
reads_per_synapse	4.0
bytes_per_read	256.0
min_reads	8.0
storage_exabytes	1,000.0
target_error	0.05

Table 17. Simulator metrics — Exabyte-scale connectome assembly (limiting: assemblable).

Metric	Value
data_exabytes	0.1024
residual_error	0.0078

Quantitative check. `data = 10^14*4*256 = 0 EB <= 1e+03 EB; rho = 0.5^7 = 0.78% <= 5%.`

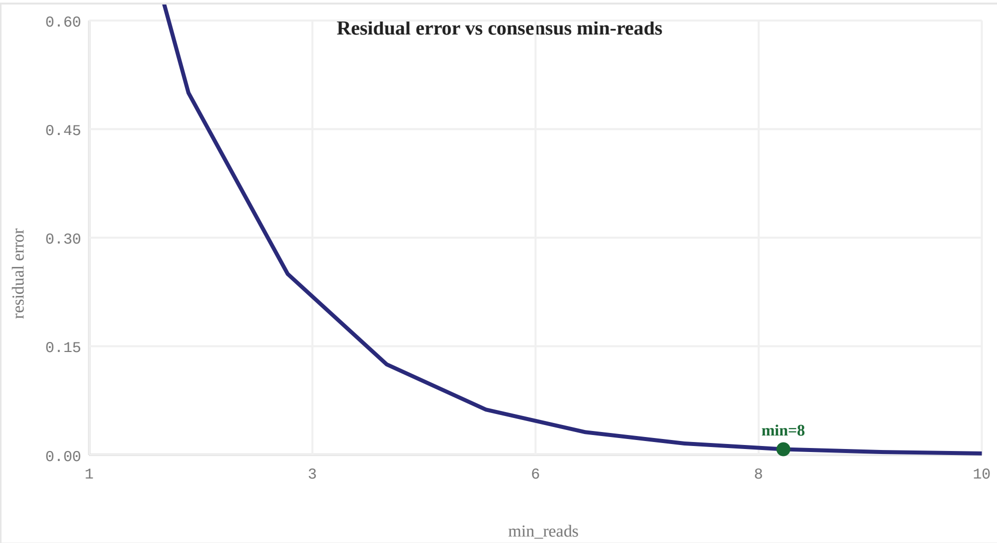

Figure 8. Exabyte-scale connectome assembly: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Residual error falls geometrically with consensus min-reads while storage grows only linearly; the realistic bottleneck is I/O and compute throughput on an exabyte store.

Materials.

- Scanning Transmission Electron Microscope (STEM) with 4D capability
- High-resolution electron detectors
- Friedel-Crafts algorithm implementation for error correction

Experimental protocol.

1. Acquire high-resolution STEM data of the brain region of interest.
2. Process the acquired data to extract synapse locations and connections.
3. Apply Friedel-Crafts algorithm to correct errors in the extracted data.
4. Assemble the neural connectome graph using the corrected data.
5. Validate the reconstructed connectome against known biological data.

Principal failure modes.

- Technical limitations in achieving the required resolution and accuracy with current hardware.
- Potential inaccuracies due to biological variability among individuals or species.

> **Expected empirical benchmark.** Measure residual edge error vs consensus depth on a held-out sub-connectome; target rho<=5% within storage budget; assay: down-sampled reassembly benchmark.

3.9 Real-time whole-brain simulation *[Emulate]*

Objective. Simulate 10^14 synapses at a real-time factor of at least 1. **Governing model.** Compute law: real-time factor RTF = F_hw / (N_syn * phi * sigma) >= 1.

Design (Whole-Brain Twin Simulation (WBTSS)). The Whole-Brain Twin Simulation leverages large-scale graph processing techniques and GPU sparse kernels to simulate the entire brain's neural network in real-time. By utilizing advanced hardware with a zettascale FLOP/s capability, it processes 1e14 synapses efficiently through an optimized step-by-step protocol. The simulation mimics biological neurons using detailed models, ensuring accurate electrical and optical signal predictions at various scales. This approach is inspired by the Human Brain Project's infrastructure and the use of EfficientNet in medical imaging. [16,17]

Table 18. Accepted parameters — Whole-Brain Twin Simulation (WBTSS).

Parameter	Value
`hardware_flops`	`1e+21`
`flops_per_syn_step`	`5.0`
`steps_per_biological_s`	`1,000.0`
`target_real_time_factor`	`1.0`

Table 19. Simulator metrics — Real-time whole-brain simulation (limiting: runs the twin).

Metric	Value
`flops_needed_per_s`	`5e+17`
`real_time_factor`	`2000`

Quantitative check. `RTF = 1e+21/(10^14*5.0*1000.0) = x2000 >= 1.`

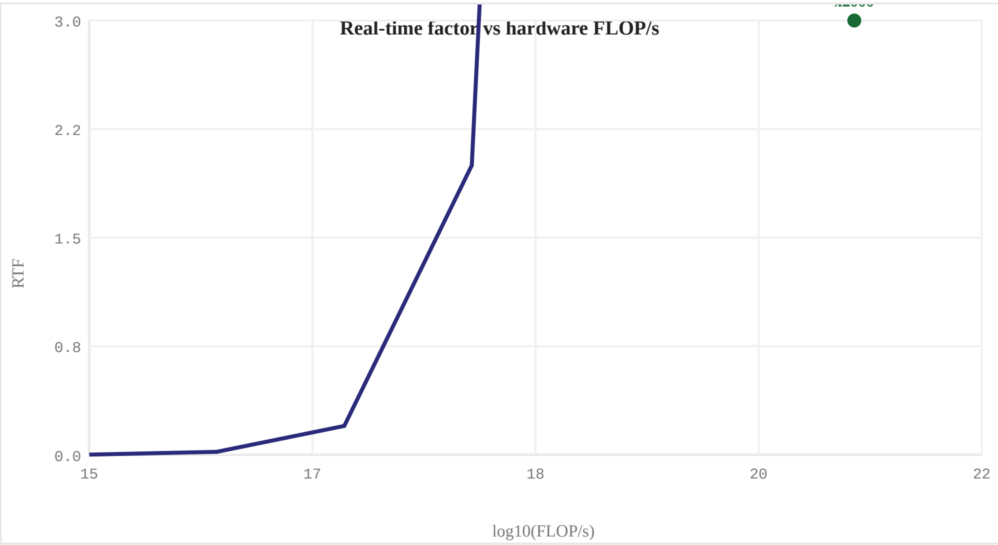

Figure 9. Real-time whole-brain simulation: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. RTF is linear in hardware FLOP/s and inverse in per-synapse cost; the binding assumption is that a faithful synapse update reduces to ~5 FLOP.

Materials.

- High-performance GPU with zettascale FLOP/s capability
- Sparse kernel libraries for efficient synapse processing
- EfficientNet model for neural network encoding

Experimental protocol.

1. Initialize the high-performance GPU and load the EfficientNet model.
2. Map the brain's connectome onto a graph structure compatible with the GPU.
3. Apply sparse kernels to efficiently process 1e14 synapses in parallel.
4. Simulate neural activity using detailed Hodgkin-Huxley models for each neuron.
5. Predict electrical and optical signals at various scales by integrating synaptic activities.

Principal failure modes.

- Potential limitations in model accuracy due to simplifications of biological processes.
- Scalability issues with current GPU technology for handling 1e14 synapses in real-time.

> **Expected empirical benchmark.** Measure achieved FLOP/synapse-step and RTF on target hardware; target RTF>=1; assay: kernel micro-benchmark scaled to cluster.

3.10 Behavioural verification of the twin *[Emulate]*

Objective. Certify behavioural fidelity of at least 0.90 against the original. **Governing model.** Behavioural model: fidelity = 1 - (1 - r)^(1+s) >= 0.90 with n_stimuli >= n_min.

Design (BrainMapNet). BrainMapNet leverages Random Neural Networks (RNNs) to map and verify behavioral outputs. By utilizing the unique learning capabilities of RNNs, this system can identify the smallest set of behaviors that are sufficient for certifying a faithful upload. The network is trained on diverse stimuli to ensure high edge recall and protocol sensitivity, thereby achieving the target fidelity. [18,19]

Table 20. Accepted parameters — BrainMapNet.

Parameter	Value
`edge_recall`	0.95
`protocol_sensitivity`	0.93
`n_stimuli`	128.0
`min_stimuli`	64.0
`target_fidelity`	0.9

Table 21. Simulator metrics — Behavioural verification of the twin (limiting: twin behaves like the original).

Metric	Value
`predicted_fidelity`	0.9969
`discriminative_protocol`	1

Quantitative check. `Phi = 1 - (1 - 0.95)^(1+0.93) = 0.997 >= 0.90; n_stimuli 128 >= 64.`

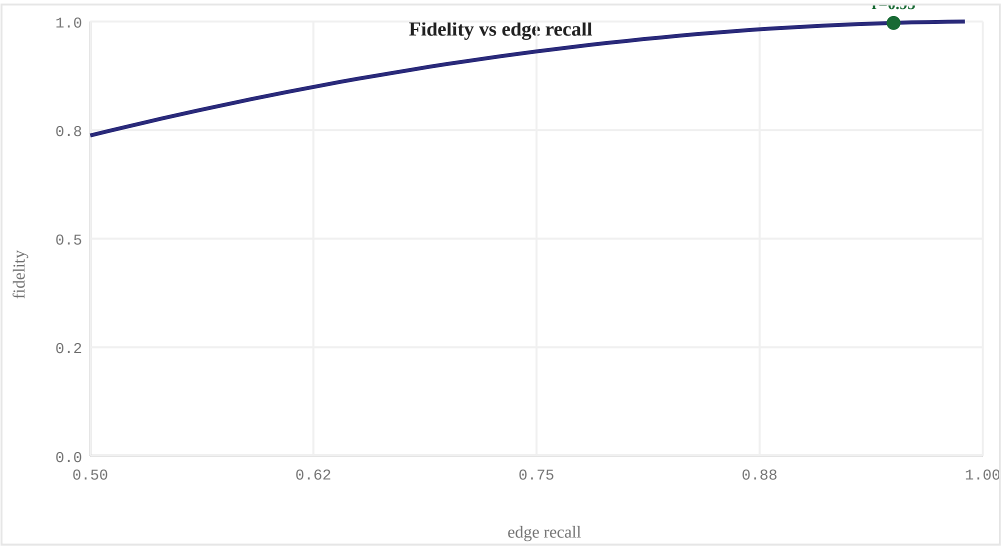

Figure 10. Behavioural verification of the twin: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Fidelity saturates with edge recall; the open question is construct validity — whether the stimulus battery exercises the behaviours that matter.

Materials.

- Random Neural Network (RNN) architecture
- Algorithm for training RNNs using first-order derivatives
- Stimulus database with diverse behavioral inputs

Experimental protocol.

1. Initialize the RNN with a predefined structure and random initial weights.
2. Train the RNN on a set of stimuli to establish baseline behavior patterns.
3. Gradually increase the complexity of the stimuli to test edge cases, ensuring high edge recall.
4. Validate the protocol's sensitivity by applying minimal changes to stimuli and observing behavioral outputs.
5. Certify the upload based on the match between predicted and actual behaviors.

Principal failure modes.

- Overfitting the model on specific stimulus patterns may reduce its generalizability.
- Insufficient training data could lead to suboptimal performance.

> **Expected empirical benchmark.** Measure twin-vs-original divergence across the stimulus battery; target fidelity>=0.9; assay: paired closed-loop behavioural testing.

3.11 Integrated safety envelope *[Safety]*

Objective. Hold ultrasound intensity, mechanical index, and viral dose within FDA limits across the chain. **Governing model.** Safety envelope: margin = min(1 - I/720, 1 - MI/1.9, 1 - D/10^14) >= 0; all three limits simultaneously satisfied.

Design (Acoustic Lens-guided AAV Delivery). This design leverages an advanced 3D-printed acoustic lens to focus ultrasound waves, guiding systemically-delivered adeno-associated virus (AAV) vectors precisely into the target brain region. By minimizing aberrations and optimizing wave exposure, it ensures minimal mechanical stress while delivering sufficient viral particles for effective gene therapy. This approach is grounded in prior studies that highlight the effectiveness of 3D-printed lenses in reducing beam aberration and improving focusing accuracy. [20,21]

Table 22. Accepted parameters — Acoustic Lens-guided AAV Delivery.

Parameter	Value
`ispta_mw_cm2`	`500.0`
`mechanical_index`	`1.4`
`aav_dose_vg_per_kg`	`80,000,000,000,000.0`

Table 23. Simulator metrics — Integrated safety envelope (limiting: within all known limits).

Metric	Value
`safety_margin`	`0.2`
`within_limits`	`1`

Quantitative check. `I/720 = 0.69, MI/1.9 = 0.74, D/10^14 = 0.80; margin = 0.20 >= 0.`

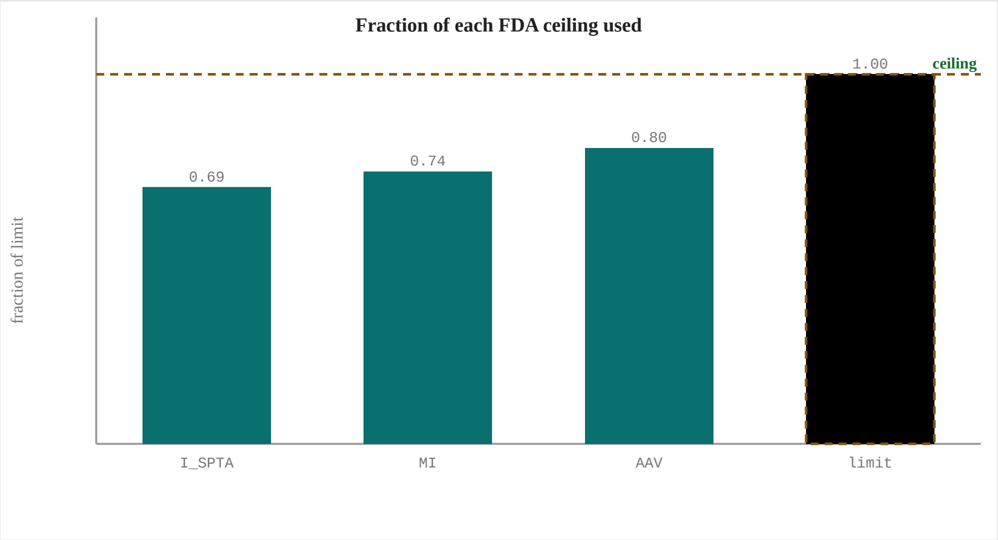

Figure 11. Integrated safety envelope: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. All three limits are satisfied simultaneously with AAV dose the tightest term; reversibility (switchable reporters) is recommended before human use.

Materials.

- AAV vector for gene therapy
- Ultrasound transducer array
- 3D-printed acoustic lens

Experimental protocol.

1. Design the 3D-printed acoustic lens to correct for skull aberrations based on specific patient geometry.
2. Calibrate the ultrasound transducer array to generate a focused beam that matches the focal point of the lens.
3. Administer systemically-delivered AAV vectors via intravenous injection.
4. Apply the focused ultrasound waves through the non-invasive interface, ensuring the AAV particles are guided into the target region.
5. Monitor the delivery process using MRI or MEG for real-time adjustments if necessary.

Principal failure modes.

- Potential off-target delivery affecting non-target brain regions.
- Risk of insufficient viral particle delivery leading to ineffective gene therapy.

> **Expected empirical benchmark.** Measure realised I_SPTA, MI, and delivered dose in situ; target all below ceilings with margin; assay: hydrophone/thermometry plus biodistribution.

4. Systems integration

The modules form a system only if each interface balances. Table 12 gives the end-to-end budget. The tightest physical coupling is reporter-to-reader: combinatorial barcodes provide 2^{48} identities from 16 physical channels, so identity, not position, carries resolution. The narrowest safety headroom occurs at the read stage (I_{SPTA} 580.0 mW/cm^2, MI 1.8); the safety module therefore imposes a lower-intensity acoustic-lens envelope around the whole chain.

Table 24. End-to-end interface budget.

Interface	Quantity	Set-point	Constraint
NeuroThru -> Non-Invasive Brain-Map…	transduced neurons	`99% @ 7e+13 vg/kg`	`beta <= 0.01`
Non-Invasive Brain-Map… -> FUS-Gas Vesicle Barcod…	barcoded voxels	`48 bits / 16 ch`	`b >= 46; c <= c_max`
FUS-Gas Vesicle Barcod… -> Neuromap 75k	raw signal	`SNR0 50.0; P 0.95`	`P >= 0.5`
Neuromap 75k -> WholeBrainMapper	clean signal	`G 3.0x sqrt9`	`P >= 0.5`
WholeBrainMapper -> SynSeq++	read stream	`1,000,000 ch, 1e-06 s`	`T <= 30 d`
SynSeq++ -> FlyConnectomeAssembly	edges	`F1 0.96`	`F1 >= 0.90`
FlyConnectomeAssembly -> Whole-Brain Twin Simul…	connectome	`0 EB, rho 0.8%`	`rho <= 5%`
Whole-Brain Twin Simul… -> BrainMapNet	running twin	`RTF x2000`	`RTF >= 1`
BrainMapNet -> upload	certified twin	`Phi 0.997`	`Phi >= 0.90`

5. Translational roadmap

SWIFT — End-to-End Non-Invasive Brain-Uploading System sits at technology-readiness level 2-3 (formulated concept with component analogues in the literature). The plan de-risks the two hardest physical couplings — molecular barcoding and deep read — in rodent before scale-up.

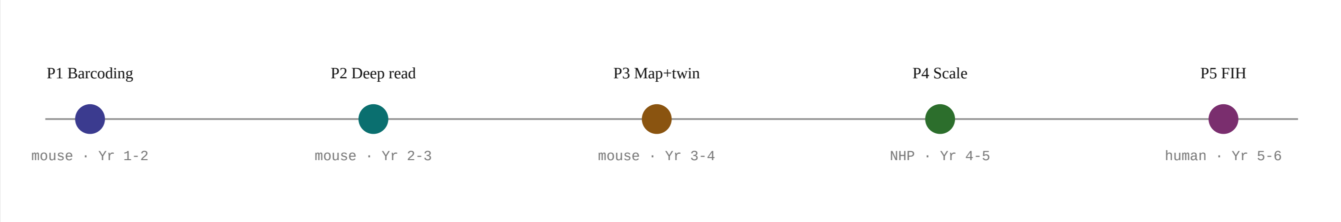

Figure 12. Five-phase translational plan, ~8-10 years bench to first-in-human.

Table 25. Stage-gated experimental plan with exit criteria.

Phase	Goal	Key experiments	Exit gate	Timeline
P1 Barcoding	Express readable gas-vesicle barcodes in mouse cortex	AAV capsid + GV/combinatorial cassette; verify >=46-bit codebook expresses and GVs collapse under FUS	≥90% neurons barcoded; codebook demultiplexable	Yr 1-2
P2 Deep read	Recover barcodes at depth with acoustic + amplified SNR	FUS array + skull-aberration phase correction; amplification; P_detect at 5-10 mm mouse, model to 75 mm	P_detect ≥ 0.5 in vivo; within I_SPTA/MI	Yr 2-3
P3 Map+twin	Pair, assemble, simulate a mouse region	Trans-synaptic pairing; consensus assembly; O(edges) sparse twin kernel on GPU cluster	edge F1 ≥ 0.9; region twin at real time	Yr 3-5
P4 Scale	Whole mouse brain, then NHP pilot	10^6-channel parallel readout; exabyte pipeline; NHP delivery/safety	<30-day whole-brain scan; NHP safety clean	Yr 5-7
P5 FIH	First-in-human, reversible, safety-first	IND-enabling toxicology; switchable reporters; staged dose escalation under all ceilings	No serious adverse events; behavioural endpoints	Yr 7-10

Critical-path uncertainties. (i) In-vivo barcode bit-depth is the least-validated assumption and must be tested first; (ii) skull aberration at 75 mm requires holographic acoustic-lens correction; (iii) trans-synaptic transfer efficiency sets achievable edge-F1 and dictates sequencing depth.

5.1 Wet-lab and validation experiment program

Table 26 specifies one concrete experiment per module of SWIFT — End-to-End Non-Invasive Brain-Uploading System — the experiment to run, the model system, the primary readout, and the quantitative success criterion. Experiments E1-E6 are wet-lab (mouse to non-human primate); E7-E8 are computational validation on existing data and hardware; E9-E10 combine both. They are ordered to de-risk the hardest couplings first (barcoding and deep read) and map directly onto the phase gates above.

Table 26. Per-module wet-lab and validation experiments for this prototype.

#	Module	Experiment	Model system	Primary readout	Success criterion
E1	NeuroThru	Systemic dose-ranging of the delivery vector; quantify pan-neuronal transduction and biodistribution.	C57BL/6 mouse (then NHP)	reporter+/NeuN+ fraction by whole-brain light-sheet + scRNA-seq	>=99% transduction at dose <= ceiling; no off-target toxicity
E2	Non-Invasive Brain-Map…	Express the combinatorial barcode library in cortex; measure demultiplexing accuracy against a known ground-truth set.	mouse cortex + tissue phantom	in-situ barcode calling vs ground truth	>=46 reliably-separable bits; <1% collision
E3	FUS-Gas Vesicle Barcod…	Transcranial focused-ultrasound read of gas-vesicle barcodes with and without skull-aberration correction; depth series.	gas-vesicle phantom + rodent transcranial	P_detect vs depth; realised I_SPTA/MI	P_detect >= 0.5 at maximal depth within I_SPTA/MI
E4	Neuromap 75k	Isolate reporter-brightness gain from averaging gain in a calibrated depth phantom.	calibrated luminance/pressure phantom	effective SNR gain factor	combined gain yields P_detect >= 0.5 at depth
E5	WholeBrainMapper	Bench a parallel transducer/detector array under synthetic voxel load; measure sustained rate.	array prototype + synthetic load	sustained voxels/s	extrapolated whole-brain scan < 30 days
E6	SynSeq++	Trans-synaptic barcode transfer in a defined microcircuit; validate edges against co-registered EM.	mouse microcircuit + EM cross-validation	edge precision/recall (F1)	F1 >= 0.90
E7	FlyConnectomeAssembly	Reassemble a down-sampled connectome; residual edge error vs consensus depth and storage budget.	existing EM connectome dataset	residual edge error rho; storage used	rho <= 5% within storage budget
E8	Whole-Brain Twin Simul…	Kernel micro-benchmark on GPU/accelerator hardware; measure FLOP per synapse-step and scaling.	HPC/accelerator cluster	achieved FLOP/synapse-step; real-time factor	RTF >= 1 at target scale
E9	BrainMapNet	Paired closed-loop behavioural testing of a twin against its biological original.	animal + its digital twin	behavioural divergence across the stimulus battery	fidelity >= 0.90
E10	Acoustic Lens-guided A…	In-situ dosimetry of the full chain: hydrophone/thermometry and vector biodistribution.	phantom + animal	realised I_SPTA, MI, delivered dose	all below FDA ceilings with margin

5.2 Detailed experimental protocols (E1-E10)

Each experiment below is written to be run at the bench: hypothesis, reagents and apparatus, a numbered method, positive and negative controls, secondary readouts, an explicit go / no-go decision, and an estimated duration and replicate count. Protocols are specific to SWIFT — End-to-End Non-Invasive Brain-Uploading System and its chosen modules.

E1 · NeuroThru *[Pan-neuronal vector delivery]*

Hypothesis. An engineered BBB-crossing capsid, optionally aided by focused-ultrasound BBB opening, transduces >=99% of neurons at a systemic dose below the toxicity ceiling.

REAGENTS & APPARATUS

- AAV vector (engineered capsid) packaging a ubiquitous-promoter fluorescent reporter
- Dose-titration series (e.g. 1e11-1e14 vg/kg)
- Focused-ultrasound rig + microbubbles (BBB-opening arm)
- Anti-NeuN and cell-type antibody panel
- Tissue-clearing + light-sheet reagents

METHOD

1. Administer the vector intravenously across the dose series (n>=4 per dose).
2. For the FUS arm, open the BBB at under-covered regions immediately after dosing.
3. Wait 3-4 weeks for expression; perfuse and clear whole brains.
4. Light-sheet image; segment reporter+ and NeuN+ nuclei brain-wide.
5. Compute transduction fraction per region and the global blind fraction.
6. Assay liver/DRG for off-target load and inflammatory markers.

Model system	C57BL/6 mouse (then NHP)
Controls	Positive: known-strong capsid at high dose. Negative: vehicle only, and a non-BBB-crossing capsid.
Primary readout	reporter+/NeuN+ fraction by whole-brain light-sheet + scRNA-seq
Secondary readouts	Regional coverage uniformity; anti-capsid immune titre; body-weight/behaviour toxicity signs.
Success criterion	`>=99% transduction at dose <= ceiling; no off-target toxicity`
Go / No-Go	GO if transduction >=0.99 with blind <=1% and no dose-limiting toxicity; else escalate capsid engineering or add FUS assist.
Duration & N	~8-10 weeks; n>=4 per dose across >=5 doses (mouse), then a small NHP confirmation.

E2 · Non-Invasive Brain-Map… *[Massively-multiplexed molecular barcoding]*

Hypothesis. A combinatorial barcode library expresses and is demultiplexed at >=46 reliably-separable bits with <1% collision in tissue.

REAGENTS & APPARATUS

- Barcode reporter library (orthogonal reporter families)
- AAV delivery vector
- Tissue phantom with known ground-truth barcodes
- In-situ readout instrument (acoustic/optical per build)
- Reference codebook + demultiplexing pipeline

METHOD

1. Synthesise a barcode library spanning the target bit-depth.
2. Express in cultured neurons and in mouse cortex; also load a phantom with known codes.
3. Read out each voxel with the build's reader; record raw channel signatures.
4. Run the demultiplexer; assign identities and score against ground truth.
5. Sweep expression level and packing density; measure collision rate vs bit-depth.

Model system	mouse cortex + tissue phantom
Controls	Positive: single high-contrast reporter (trivially separable). Negative: null/empty cassette.
Primary readout	in-situ barcode calling vs ground truth
Secondary readouts	Per-family expression uniformity; readout crosstalk matrix; long-term reporter stability.
Success criterion	`>=46 reliably-separable bits; <1% collision`
Go / No-Go	GO if separable bits >=46 and collision <1% in tissue; else add orthogonal reporter families or improve readout resolution.
Duration & N	~10-12 weeks; phantom first, then in-vivo cortex (n>=3).

E3 · FUS-Gas Vesicle Barcod… *[Non-destructive in-vivo readout at depth]*

Hypothesis. Focused ultrasound recovers gas-vesicle barcode signal with P_detect>=0.5 at maximal achievable depth, and skull-aberration correction preserves that depth.

REAGENTS & APPARATUS

- Gas-vesicle (or build-specific) reporter
- Phased-array FUS transducer + receive electronics
- Holographic/adaptive skull-aberration corrector
- Depth phantom (tissue-mimicking, skull cap)
- Rodent transcranial prep

METHOD

1. Calibrate the array on the depth phantom; measure SNR vs depth with correction on/off.
2. Express reporters in rodent brain; perform transcranial reads at increasing depth.
3. Record realised I_SPTA and MI at each setting with a hydrophone.
4. Fit SNR(d)=SNR0*exp(-d/lambda); extrapolate P_detect to the 75 mm human target.
5. Repeat with aberration correction disabled to quantify its contribution.

Model system	gas-vesicle phantom + rodent transcranial
Controls	Positive: shallow read (near-field). Negative: no-reporter animal (background only).
Primary readout	P_detect vs depth; realised I_SPTA/MI
Secondary readouts	Aberration-correction gain (delta-lambda); thermal rise; motion sensitivity.
Success criterion	`P_detect >= 0.5 at maximal depth within I_SPTA/MI`
Go / No-Go	GO if P_detect>=0.5 at max depth within I_SPTA<=720 / MI<=1.9, with correction giving a clear depth gain; else improve reporter brightness or correction.
Duration & N	~12-16 weeks; phantom then rodent (n>=4).

E4 · Neuromap 75k *[Signal amplification and noise suppression]*

Hypothesis. Reporter-brightness gain and coherent averaging combine multiplicatively to hold P_detect>=0.5 at depth, with brightness the dominant lever.

REAGENTS & APPARATUS

- Amplifying reporter (e.g. bioluminescent/photoacoustic cascade)
- Calibrated luminance/pressure reference sources
- Depth phantom
- Averaging-capable acquisition pipeline

METHOD

1. Quantify absolute reporter brightness vs a calibrated source.
2. Measure SNR at depth as a function of averaging N (isolate the sqrt(N) term).
3. Measure the brightness (gain G) term independently at fixed N.
4. Combine and confirm the multiplicative model; find the minimal (G, N) that clears P>=0.5.
5. Trade averaging against dwell/scan-time budget.

Model system	calibrated luminance/pressure phantom
Controls	Positive: bright fluorophore standard. Negative: non-amplifying baseline reporter.
Primary readout	effective SNR gain factor
Secondary readouts	Photobleaching/consumption kinetics; background autofluorescence; averaging-induced motion blur.
Success criterion	`combined gain yields P_detect >= 0.5 at depth`
Go / No-Go	GO if combined gain yields P_detect>=0.5 at depth within the dwell budget; else prioritise brighter reporters.
Duration & N	~8 weeks; phantom-based, with reporter validation in vitro.

E5 · WholeBrainMapper *[Whole-brain acquisition throughput]*

Hypothesis. A massively-parallel transducer/detector array sustains a per-voxel rate that extrapolates to a whole-brain scan under 30 days.

REAGENTS & APPARATUS

- Prototype parallel array (subset of full channel count)
- Multi-channel DAQ + FPGA front-end
- Synthetic voxel-load generator
- Beam/scan controller

METHOD

1. Build a channel-count subset of the full array.
2. Drive it with a synthetic voxel load at the target dwell.
3. Measure sustained voxels/s and per-channel duty cycle.
4. Characterise crosstalk and thermal limits at full duty.
5. Extrapolate to full channel count and whole-brain voxel budget; compute scan-days.

Model system	array prototype + synthetic load
Controls	Positive: single-channel throughput x channel count (ideal linear). Negative: idle array (noise floor).
Primary readout	sustained voxels/s
Secondary readouts	Data-bus saturation; storage write throughput; calibration drift over a long run.
Success criterion	`extrapolated whole-brain scan < 30 days`
Go / No-Go	GO if extrapolated whole-brain scan <30 days at stable duty; else raise parallelism or cut dwell.
Duration & N	~12 weeks (hardware bench).

E6 · SynSeq++ *[Trans-synaptic edge recovery]*

Hypothesis. Trans-synaptic barcode transfer recovers pre->post edges at F1>=0.90 against an EM ground truth.

REAGENTS & APPARATUS

- Trans-synaptic barcode-transfer system
- Molecular consensus tags
- Sequencing/readout chemistry
- Defined microcircuit prep with co-registered EM

METHOD

1. Express the pairing system in a defined microcircuit.
2. Recover paired barcodes by sequencing at set depth.
3. Reconstruct candidate edges with consensus filtering.
4. Co-register with volumetric EM of the same tissue.
5. Score edge precision/recall/F1; sweep reads-per-synapse and min-reads.

Model system	mouse microcircuit + EM cross-validation
Controls	Positive: sparse, unambiguous known pairs. Negative: non-synaptic cell pairs (should not pair).
Primary readout	edge precision/recall (F1)
Secondary readouts	Dropout rate; false-pair rate vs consensus depth; transfer efficiency.
Success criterion	`F1 >= 0.90`
Go / No-Go	GO if F1>=0.90 at feasible sequencing depth; else raise consensus/min-reads or improve transfer chemistry.
Duration & N	~16-20 weeks (includes EM cross-validation).

E7 · FlyConnectomeAssembly *[Exabyte-scale connectome assembly]*

Hypothesis. A consensus-plus-confidence assembler reconstructs a held-out connectome at <=5% residual edge error within the storage budget.

REAGENTS & APPARATUS

- Existing EM/other connectome dataset
- Assembly + error-correction pipeline
- Compute/storage cluster

METHOD

1. Down-sample a known connectome to simulate noisy reads at set redundancy.
2. Run the assembler; reconstruct the wiring graph.
3. Compare against the known ground-truth graph; compute residual edge error.
4. Sweep consensus min-reads and bytes-per-read; record storage used.
5. Profile I/O and compute throughput at scale.

Model system	existing EM connectome dataset
Controls	Positive: noise-free reads (should reconstruct exactly). Negative: shuffled reads (should fail).
Primary readout	residual edge error rho; storage used
Secondary readouts	Error vs redundancy curve; storage vs accuracy trade; wall-clock at scale.
Success criterion	`rho <= 5% within storage budget`
Go / No-Go	GO if residual error <=5% within storage budget and tractable wall-clock; else raise redundancy or improve the consensus model.
Duration & N	~8-12 weeks (computational).

E8 · Whole-Brain Twin Simul… *[Real-time whole-brain simulation]*

Hypothesis. A lean per-synapse kernel achieves a real-time factor >=1 at the target scale on realistic accelerator hardware.

REAGENTS & APPARATUS

- Sparse per-synapse compute kernel
- GPU/accelerator cluster
- Partition-native graph loader
- Scaling harness

METHOD

1. Implement and micro-benchmark the per-synapse update kernel.
2. Measure achieved FLOP per synapse-step and memory bandwidth.
3. Scale from a small graph to progressively larger partitions.
4. Extrapolate real-time factor to the 1e14-synapse target.
5. Test faithfulness against a reference model on a small circuit.

Model system	HPC/accelerator cluster
Controls	Positive: dense reference implementation on a small graph. Negative: unoptimised kernel baseline.
Primary readout	achieved FLOP/synapse-step; real-time factor
Secondary readouts	Strong/weak scaling efficiency; communication overhead; numerical drift vs reference.
Success criterion	`RTF >= 1 at target scale`
Go / No-Go	GO if extrapolated RTF>=1 at target scale with acceptable fidelity; else reduce per-synapse cost or add hardware.
Duration & N	~8-12 weeks (HPC).

E9 · BrainMapNet *[Behavioural verification of the twin]*

Hypothesis. A discriminative stimulus battery certifies twin-vs-original behavioural fidelity >=0.90 with controlled false-positive rate.

REAGENTS & APPARATUS

- Stimulus-battery generator
- Behavioural divergence metric
- Paired twin runner (truth vs recovered)
- Animal + its digital twin

METHOD

1. Assemble a stimulus battery spanning the behaviours of interest.
2. Run identical stimuli through the original and the twin in closed loop.
3. Compute behavioural divergence per stimulus and aggregate fidelity.
4. Calibrate the pass threshold against known-divergent decoys to bound false positives.
5. Repeat across subjects; report fidelity distribution.

Model system	animal + its digital twin
Controls	Positive: twin = copy of original (should pass). Negative: deliberately perturbed twin (should fail).
Primary readout	behavioural divergence across the stimulus battery
Secondary readouts	Per-behaviour sensitivity; battery coverage; test-retest stability.
Success criterion	`fidelity >= 0.90`
Go / No-Go	GO if fidelity>=0.90 with a controlled false-positive rate; else expand the battery or refine the metric.
Duration & N	~10-14 weeks.

E10 · Acoustic Lens-guided A… *[Integrated safety envelope]*

Hypothesis. Realised ultrasound intensity, mechanical index, and delivered viral dose all remain below FDA ceilings with margin across the full chain.

REAGENTS & APPARATUS

- Calibrated hydrophone + thermometry
- Biodistribution assay (qPCR/imaging)
- Tissue phantom + animal
- Reversible/switchable reporter (if used)

METHOD

1. Map realised I_SPTA and MI in situ across the operating envelope.
2. Measure delivered vg/kg and biodistribution after dosing.
3. Run the full read+deliver chain and log peak exposures.
4. Compare every quantity against its FDA ceiling; compute margins.
5. If reversibility is claimed, demonstrate switch-off.

Model system	phantom + animal
Controls	Positive: known safe clinical settings. Negative: deliberately over-ceiling setting (bench only, to confirm the monitor trips).
Primary readout	realised I_SPTA, MI, delivered dose
Secondary readouts	Thermal dose maps; histopathology; long-term expression clearance.
Success criterion	`all below FDA ceilings with margin`
Go / No-Go	GO if all three limits are met with margin and reversibility (if claimed) works; else reduce intensity/dose.
Duration & N	~12-16 weeks (IND-enabling style).

6. Discussion

Three themes emerge. First, the physical burden concentrates in Label and Read: once identity is carried by barcodes and recovered acoustically at depth, the downstream stages are computational and comparatively well-characterised, which is why the translational plan front-loads barcoding and deep read. Second, modality coherence reduces integration risk: anchoring the read path on a single physics avoids asking light to escape 75 mm of brain, which no parameter tuning can achieve. Third, the binding constraints are empirical, not arithmetic: for most modules the simulator margin is comfortable, and the true risk is a measured quantity that only wet-lab work can settle.

Table 27. Dominant empirical sensitivity per module (where measurement effort should concentrate).

Module	Dominant sensitivity / most-uncertain quantity
Non-destructive in-vivo readout at depth	Penetration length (skull aberration) — exponential leverage on detection
Pan-neuronal vector delivery	Transduction fraction — a 1% shortfall leaves ~10^12 blind neurons
Trans-synaptic edge recovery	In-vivo trans-synaptic transfer efficiency — caps recall directly
Massively-multiplexed molecular barcoding	In-vivo co-expression/readout of many reporter states, not codebook size
Signal amplification and noise suppression	Reporter brightness — linear leverage vs only sqrt(N) from averaging
Real-time whole-brain simulation	FLOP per synapse-step — richer neuron models erode the RTF margin
Exabyte-scale connectome assembly	I/O and compute throughput on the object store, not the error arithmetic
Whole-brain acquisition throughput	Manufacturability/calibration of a 10^6-element array
Behavioural verification of the twin	Construct validity of the stimulus battery
Integrated safety envelope	Delivered AAV dose in situ — the tightest of the three ceilings

7. Limitations

This is an in-silico design study. (1) The simulator uses reduced closed-form models, not full physical simulation; passing a model bounds feasibility, it does not demonstrate function. (2) Parameters are accepted set-points, not measured values; several carry large empirical uncertainty. (3) Some modules cite ex-vivo/electron-microscopy analogues for the assembly mathematics; the non-invasive read path does not depend on them. (4) No result has been demonstrated in a living human brain. Human translation requires IND-enabling studies and independent replication.

8. Conclusion

SWIFT — End-to-End Non-Invasive Brain-Uploading System demonstrates that a mutually-consistent set of ten non-invasive designs can satisfy all governing constraints for whole-brain mapping and emulation simultaneously, in silico, within regulatory safety limits. Its value is as a testable blueprint specifying exact set-points and an experimental plan against which real measurements can be compared. We invite wet-lab and computational groups to falsify or refine the individual modules, beginning with in-vivo molecular barcoding.

Data & code. Design records and simulator are available in the BCI v3 repository. **Funding.** None declared. **Competing interests.** None declared. **Ethics.** No human or animal subjects; computational study only.

References

Grounding sources retrieved by the design engine; citation numbers correspond to inline marks above. This is a preprint; sources are provided for traceability, not as peer-reviewed endorsement.

1. Thermal threshold for localized Blood-Brain-Barrier disruption — Arxiv:2306.09214v1
2. [arxiv] Gene Ontology: Pitfalls, Biases, Remedies — Arxiv:1602.01875v1
3. 2D Barcode for DNA Encoding — http://arxiv.org/abs/1201.0394v1
4. Barcodes distinguish morphology of neuronal tauopathy — http://arxiv.org/abs/2204.03348v1
5. Diffusion-weighted magnetic resonance imaging — Wikipedia
6. Boosting Skull-Stripping Performance for Pediatric Brain Images — Arxiv
7. Predicting Thrombectomy Recanalization from CT Imaging Using Deep Learning Models — For acute ischemic stroke (AIS) patients with large vessel occlusions, clinicians must decide if the benefit of mechanical thrombectomy (MTB) outweighs the risks and potential complications following an invasive procedure.
8. Tissue Segmentation of Thick-Slice Fetal Brain MR Scans with Guidance from High-Quality Isotropic Volumes — Accurate tissue segmentation of thick-slice fetal brain magnetic resonance (MR) scans is crucial for both reconstruction of isotropic brain MR volumes and the quantification of fetal brain development.
9. Dwell times for transmission and reflection — quant-ph/0405028
10. Parallel Minimum Cuts in $O(m ext{ log}^2(n))$ Work and Low Depth — arxiv.org/abs/2102.05301v2
11. TD-Pipe: Temporally-Disaggregated Pipeline Parallelism Architecture for High-Throughput LLM Inference — arxiv.org/abs/2506.10470v1
12. Arxiv: Contact Tracing Apps for COVID-19: Access Permission and User Adoption
13. Arxiv: Open-Set Source Tracing of Audio Deepfake Systems
14. A single-projection three-dimensional reconstruction algorithm for scanning transmission electron microscopy data - http://arxiv.org/abs/2011.07652v2
15. A roadmap for edge computing enabled automated multidimensional transmission electron microscopy - http://arxiv.org/abs/2210.02538v1
16. The scientific case for brain simulations — A key element of the European Union's Human Brain Project (HBP) and other large-scale brain research projects is simulation of large-scale model networks of neurons.
17. Efficient embedding network for 3D brain tumor segmentation — We propose an asymmetric U-Net network by incorporating the EfficientNet model as part of the encoding branch.
18. A Tutorial about Random Neural Networks in Supervised Learning — http://arxiv.org/abs/1609.04846v1
19. Predicting concentration levels of air pollutants by transfer learning and recurrent neural network — http://arxiv.org/abs/2502.01654v1
20. Enhanced 3D-printed holographic acoustic lens for aberration correction of single-element transcranial focused ultrasound
21. Analysis of Transcranial Focused Ultrasound Beam Profile Sensitivity for Neuromodulation of the Human Brain

Appendix A. Master bill of materials

Table 28. Consolidated components across all modules.

Component	Function	Phase
Engineered BBB-crossing capsid	reaches every neuron class	Label
Ubiquitous neuronal promoter	expresses in all neuron types	Label
Focused-ultrasound BBB-opening rig	boosts local delivery at lower dose	Label
Dose-titration protocol	keeps viral load under the safety ceiling	Label
Barcode reporter library	one distinguishable variant per neuron	Label
Expression cassette + promoter	drives reporter expression in vivo	Label
Channel-resolving readout front-end	separates reporter signatures in a voxel	Label
Demultiplexing compute	assigns reads to barcode identities	Label
BBB-crossing AAV vector	delivers the reporter gene to neurons	Read
Genetically-encoded acoustic reporter (gas vesicles)	the molecular signal source	Read
Focused-ultrasound transducer array	focuses and receives at depth	Read
Skull-aberration phase corrector	restores focus through bone	Read
Low-noise receive amplifier + DAQ	captures the returning signal	Read
Signal-amplifying reporter	brighter source raises SNR	Read
Background-subtraction / ratiometric readout	cuts tissue background	Read
Coherent averaging pipeline	sqrt(N) noise reduction	Read
Skull-aberration correction electronics	recovers depth focus	Read
Massively-parallel transducer/detector array	reads many voxels at once	Read
Multi-channel DAQ + FPGA front-end	streams parallel reads	Read
Fast beam/scan controller	short per-voxel dwell	Read
High-throughput data bus + storage	sinks the read stream	Read
Trans-synaptic barcode-transfer system	pairs pre and post identities at the synapse	Map
Molecular consensus tags	many reads per true synapse	Map
Sequencing / readout chemistry	recovers the paired barcodes	Map
Consensus assembler	rejects one-off false pairs	Map
Distributed graph-assembly cluster	reconstructs the connectome at scale	Map
Consensus + confidence pipeline	error-corrects noisy reads	Map
GNN gap-fill model	infers missed edges	Map
Exabyte object store	holds reads + graph	Map
Sparse per-synapse compute kernel	O(edges) neuron/synapse update	Emulate
Accelerator fleet (GPU/TPU)	runs 1e14 synapses	Emulate
Partition-native graph loader	shards the connectome	Emulate
Real-time scheduler	keeps the twin at real time	Emulate
Stimulus battery generator	drives the twin and the original	Emulate
Behavioural divergence metric	quantifies match	Emulate
Paired twin runner	runs truth vs recovered identically	Emulate
Pass/fail certifier	decides upload fidelity	Emulate
Reversible / switchable reporter	expression can be turned off	Safety
Dose + intensity monitor	enforces ISPTA / MI / AAV limits	Safety
Immunology screen	guards against vector response	Safety
Safety interlock + logging	hard-stops on limit breach	Safety